# Technology Stack & Pipeline



## Voice Assistant Pipeline

### W Faster-Whisper — ASR

CTranslate2-optimized Whisper model (4x faster than OpenAI).
Converts spoken English audio into text transcriptions.

- Model: whisper-small (244M params, ~2GB)
- Quantization: int8 (CPU) / float16 (CUDA GPU)
- Includes ASR corrections for German city names
- Fallback: standard OpenAI Whisper if unavailable

### G GLiNER + Regex — NLU

Hybrid NLU: regex detects intent, GLiNER extracts entities.
Zero-shot entity recognition — no training data needed.

- Intents: weather_query, create/get/update/delete appointment
- Entities: location, date, time, title, description
- spaCy pipeline for text preprocessing
- RapidFuzz for fuzzy string matching (typo tolerance)

### T Template-Based — NLG

Generates natural language responses from structured data.
Uses randomized conversational fillers for naturalness.

- Templates for weather, appointments, errors, greetings
- Dynamic slot filling (location, temp, date, time)
- Conversational fillers: "Sure!", "Let me check..."
- Context-aware responses based on dialogue history

### P Piper — TTS

Fast neural text-to-speech engine, runs fully offline.
Converts NLG text output into spoken audio.

- Voice: en_US-lessac-medium (VITS neural model)
- Auto-downloads voice models on first use (~50MB)
- Low latency: suitable for real-time conversation
- Fallback: pyttsx3 (cross-platform, no download)

# Supporting Technologies & Integration



## Supporting Technologies

| Technology | Role in System |
|---|---|
| **Python 3.11+** | Core language for all modules |
| **spaCy** | NLP preprocessing (tokenization, POS tagging) |
| **RapidFuzz** | Fuzzy matching for entity normalization |
| **SpeechRecognition** | Microphone audio capture (voice mode) |
| **PyAudio** | Low-level audio I/O for mic input |
| **Requests** | HTTP client for Weather & Calendar APIs |
| **python-dateutil** | Natural date parsing ("tomorrow", "Friday") |
| **NumPy** | Audio array processing for ASR |
| **Docker** | Containerized deployment (CPU/CUDA) |
| **CTranslate2** | Optimized inference backend for Whisper |

## External APIs

**Weather API (POST)**
Endpoint: api.responsible-nlp.net/weather.php
Returns 7-day forecast with min/max temps & conditions.
Supports condition checks: rain, snow, sunny, cloudy, etc.

**Calendar API (REST CRUD)**
Endpoint: api.responsible-nlp.net/calendar.php
Full CRUD: Create, Read, Update, Delete appointments.
Uses unique calendar_id per team to avoid conflicts.

## Dialogue Management

**Context Tracking**
Maintains up to 10 conversation turns with timestamps

**Anaphora Resolution**
Resolves "there" → previous location, "that" → last appt.

**Multi-Turn State**
Prompts for missing info (e.g., asks for date if not provided)

**Filler Removal**
Strips "um", "uh", "like", "you know" from ASR output

### Design Principles

Fully On-Device • All core models (ASR, NLU, TTS) run locally — no cloud AI services needed.
Graceful Fallbacks • Each component has a fallback: Faster-Whisper → Whisper, Piper → pyttsx3, GLiNER → Regex-only NLU.